

Project 2 – ML Construction

Updated Oct 5 (problem 2)

We’re returning to the data sets from last week. We’re going to dig a bit more into different ways we could have approached the problems. You are welcome to copy-paste-modify the helper code I gave for last project.

First, we’ll return to the *wz* data set.

1. Lets explore model complexity and a bit of interpretability.
 - (a) Set up a new classifier model that uses shallow *binary* logistic regression. You can use a decision threshold of 0.5. Make sure to adjust the loss function in the training loop accordingly.
 - (b) Train both this new classifier and the one I gave you last week on the “clean” data set. Compare how they both perform on the “corrupted” data set.
 - (c) Train your new classifier instead on the “corrupted” data set. How does it handle the “safe” data set?
 - (d) Because your new models are both *shallow*, you can actually make sense of the weight matrices they have learned. Using the weights of each model, plot the decision boundary of the models in the (x_1, x_2) plane. Guided by these plots (and/or analysis of the model weights), figure out what I did to the “corrupted” data set.
2. **Regression is not classification.** *Note, we’ll be talking about some of the concepts for this problem on Monday.* Imagine now that the data labeled “0” represent background events, and the data labeled “1” represent signal events.
 - (a) Using your “clean”-trained shallow binary regression classifier, build the ROC curve and compute the AUC. Pick a few potentially-interesting threshold choices to build the full confusion matrix and compute the MCC.
 - (b) Suppose that our physics model suggests that we will encounter 10^{-4} fewer signal events than noise events. Using your model measures, choose an operating threshold and estimate the number of signal events that actually get flagged by your model. (*I can provide an additional data set to test your estimate if you are interested*). What factors influenced your decision, and what additional factors might influence your decision for a real particle experiment?

Then, we’ll follow up on the “astro” data.

3. Experimenting with different approaches to the prediction problem.
 - (a) Build and train a new prediction pipeline which takes the spectrum as the input, performs multi-class classification, and then makes use of that classification to make an age prediction. (*I can provide some hints of ways I considered doing this*). Use your prediction pipeline to reconstruct the age $\leftrightarrow$ guiding radius relation for the test set. Compare with the relation you found last week for the train and validate set. Does your new model do better or worse? Any hypotheses as to why?
 - (b) Build and train a new prediction pipeline which now *additionally* takes the guiding radius as part of its input and outputs a predicted age. Compare with the relation you found last week for the train and validate set. Does this new model do better or worse? Any hypotheses as to why?
4. **Pre-ruminations for week 4:** Imagine that the age predictor is part of a measurement pipeline. What information about its response would you need to propagate to use its predictions for population level inference? Under what assumptions should the uncertainty from prediction errors decrease as $1/\sqrt{N}$? Does our imagined situation fit that assumption?